CS 7050 — Data Warehousing and Data Mining

Term Project Report

Kennesaw State University · College of Computing and Software Engineering

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Section	W01
Project title	From Clickstream to Early Alert: A Dimensional Warehouse and Mining Pipeline for Virtual Learning Environment Engagement Data
Code	`https://github.com/kayrandmorgan-oss/cs7050-oulad-early-alert`
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Summary. The 10.66M-row OULAD clickstream and its six companion tables were loaded into a MySQL 8 fact constellation: two facts, six conformed dimensions, and two materialized aggregates. The warehouse rebuilds from the raw CSVs with one command in 12 minutes and passes 28 of 28 reconciliation checks. In particular, all 39,605,099 clicks reach the fact table. An 11-query OLAP pack returns identical results from the base fact and the aggregates, and the aggregates answer 7 of the queries up to **688**× faster. From a week-4 feature view with a machine-verified no-leakage cutoff, logistic regression predicts fail/withdraw for students still enrolled on day 28. On held-out October presentations it reaches **ROC-AUC 0.719**, which is **+8.1** accuracy points over the majority class. At a threshold tuned on training data it recalls **80%** of at-risk students. The result is modest, and it is reported as modest. k-means finds four engagement archetypes whose fail/withdraw rates range from 29% to 65%. The disparate-impact audit passes the four-fifths screen for gender, deprivation (IMD) and disability, but not for age band. Given a schema context generated from the live warehouse, an open 7B code model answers **47.5%** of 40 held-out analyst questions exactly and chooses the aggregate tables for 91% of eligible questions. Without that context it answers none. The LoRA condition is packaged for the course GPU hub.

1. Problem and research question

Virtual learning environments (VLEs) log every click, but in tables built to run courses rather than to analyze them. This project builds both halves of CS 7050 on one public dataset. The first half is a warehouse that answers coordinator questions in seconds, such as “how does week-by-week engagement in this module compare with the same module last term, by activity type?”. The second half is an early-alert mining layer trained on feature marts drawn from that warehouse. The research question, unchanged from the proposal, is: *using only VLE interaction and registration data observable through the first four weeks of a presentation, and excluding protected attributes, how accurately can final outcome (pass/distinction vs. fail/withdrawn) be predicted, and what engagement archetypes exist?* Week 4 is chosen because an alert at that point can still be acted on. Published accuracy figures on this dataset that use whole-course features are leaky for this purpose.

2. Data and profiling

Source. The data is the Open University Learning Analytics Dataset (OULAD) [1], CC BY 4.0: 22 module presentations of 7 modules (2013–2014; “B” = February start, “J” = October start). In October 2026 the official download link redirected to a page without the archive. The CSVs were therefore taken from a public mirror, and the 8-part `studentVle` was reassembled. Every row count matches the dataset paper (Table **??**), and the reconciliation suite (Section **??**) checks the files end to end. An independent second mirror had the same clickstream row count but capped `sum_click` at 6, so it was rejected as a cleaned derivative.

Table 1: Source tables and profiling highlights (`etl/profile_raw.py`, `out/profile_report.json`).

Table	Rows	Profiling findings
`studentVle`	10,655,280	days -25..269; 2,195,960 rows repeat a (student, site, day) key; max 6,977 clicks; 39,605,099 clicks total
`studentAssessment`	173,912	173 NULL scores; 1,909 banked; 2,057 submitted before day 0
`studentInfo`	32,593	28,785 distinct students; `imd_band` NULL ×1,111 and `'10-20'` missing its % ×3,516; `age_band` varies for 72 students
`studentRegistration`	32,593	45 NULL registration dates; 5,040 unregistered before day 28; 93 Withdrawn without a date
`vle`	6,364	20 activity types; `week_from/to` 82% NULL (not used)
`assessments`	206	11 exams with no date
`courses`	22	lengths 234–269 days

Each anomaly has a documented decision (D1–D14 in `etl/DECISIONS.md`). The four that most shape the results:

- **D8.** Repeated (student, site, day) rows are *summed*, not dropped. They are separate interaction records. Dropping them would lose 20% of rows and break the click-total reconciliation.
- **D9.** Zero-activity rows are generated only inside a student's *registered span*, so a student who has left is not padded with fake inactivity.
- **D13.** The 5,040 enrollments that unregistered before day 28 stay in the warehouse but leave the modeling population. An alert cannot help a student who has already left, and keeping them would inflate accuracy with trivially easy cases.
- **D14** (found by the OLAP pack, not the profiler). In BBB, CCC and DDD, 97–100% of computer-marked assessments (CMAs) are "submitted" 2–3 days after their listed date. In FFF and GGG, all CMAs share one end-of-module date. Lateness is therefore interpreted for tutor-marked assessments (TMAs) only.

3. Warehouse design

Conceptual model. Figure **??** shows the source ER model. `studentInfo` and `studentRegistration` share the composite key (module, presentation, student). `studentVle` and `studentAssessment` are the two event streams. They sit at different grains: daily site clicks, and one row per submitted assessment.

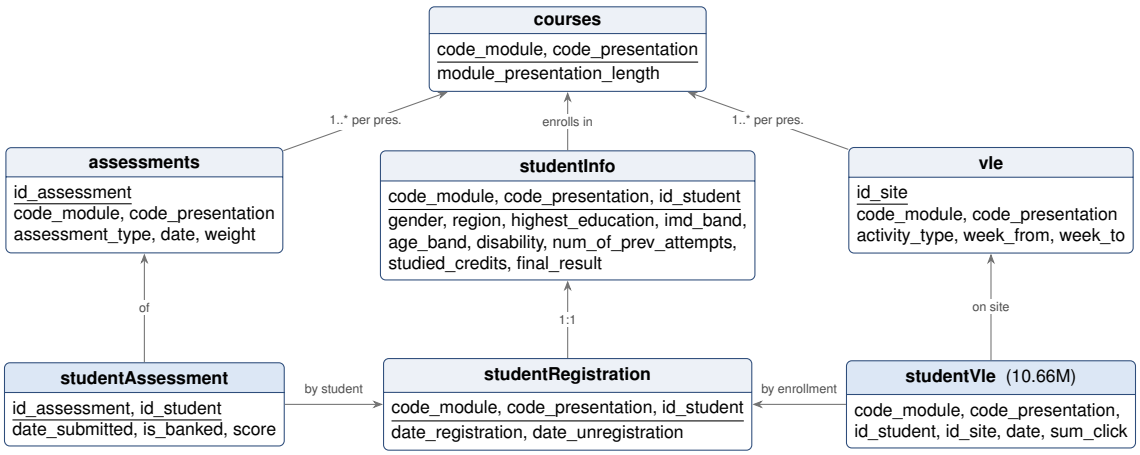


Figure 1: Conceptual ER model of the seven OULAD source tables (event streams shaded). Underlined attributes are keys.

Dimensional model. The source becomes a fact constellation of two facts that share conformed dimensions (Figure **??**), following Kimball's four-step method [3]. Every dimension uses integer surrogate keys, and natural keys are kept as degenerate attributes.

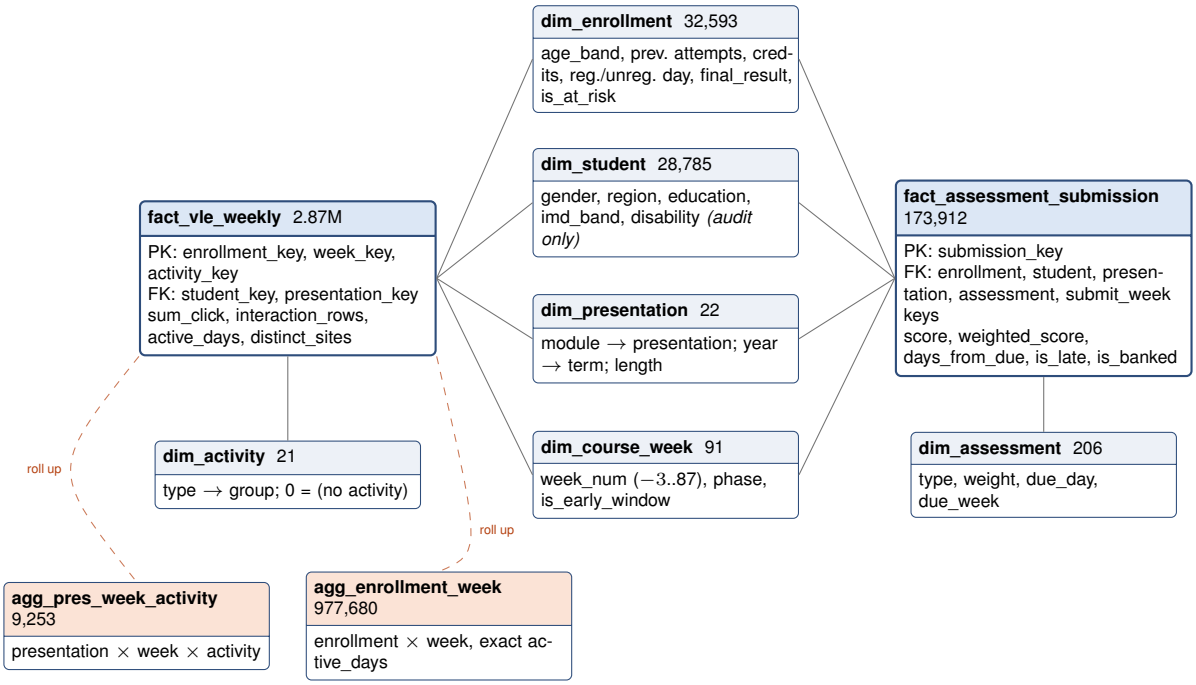


Figure 2: The fact constellation: two facts (blue), six conformed dimensions, and two materialized aggregates (orange). Row counts are after the load.

Grain decisions.

- **Weekly rather than daily.** Daily grain would reproduce the 10.66M-row source. Weekly grain matches the alert cadence and cuts the clickstream to 2.52M clicked rows.
- **Relative course week, not calendar date.** `week_num = FLOOR(day/7)+1`, so week 1 is days 0–6 and weeks ≤ 0 are pre-start. This makes presentations of different lengths and start dates directly comparable.
- **`activity_type` as a dimension, not 20 pivoted columns.** This refines the proposal's grain statement. Fact 1 is one row per *enrollment* × *course week* × *activity type with clicks*, plus exactly one `(no activity)` row (`activity_key` 0) for each registered enrollment-week with no clicks (350,649 rows). "Clicked nothing in week 3" becomes a countable fact without multiplying zero rows by 20 types. `sum_click` is fully additive. `active_days` is additive across weeks but not across activity types, which is why the aggregate `agg_enrollment_week` computes it exactly from staging.
- **`age_band` on the enrollment, other demographics on the student.** `age_band` changes between presentations for 72 students. Gender, region, education, IMD and disability never change, so they are Type-1 attributes of `dim_student`.

Table comments hold each table's grain statement. The text-to-SQL prompt (Section **??**) reads these comments from `information_schema`.

4. ETL and reconciliation

`etl/build_warehouse.py` drops and rebuilds the database in three layers.

1. **Staging.** The CSVs are loaded as-is with `pandas` and `SQLAlchemy`. The clickstream is loaded in 200k-row chunks. The proposal's `LOAD DATA INFILE` fallback is kept behind `--load-data` but was not needed.
2. **Transform and load.** The 17 timed SQL steps in `sql/transform_load.sql` cover surrogate-key assignment with `ROW_NUMBER()`, the cleaning decisions, zero-row generation, the aggregates, and foreign keys added after the bulk load.
3. **Reconciliation.** 28 assertions must all pass, or the build exits non-zero and nothing downstream runs.

Table 2: Build timings (2 vCPU, 8 GB, MySQL 8.0.46; `out/build_log_initial_run.json`).

Step	s	Step	s
Stage 10.66M clicks (chunked pandas)	267.2	Zero-activity rows (fact + A2)	21.7
Fact 1, clicked rows (GROUP BY)	130.8	A1 `agg_pres_week_activity`	11.9
A2 `agg_enrollment_week`	95.7	Foreign keys on facts	63.3
Dimensions (6)	17.8	Reconciliation (28 checks) + DB setup	94.6
Total, raw CSVs → verified warehouse			**723.8**

Table 3: Reconciliation suite (selected; all 28 pass). R00 checks compare against the CSV files themselves.

	Assertion	Expected	Actual
R00	`studentVle.csv` click total = fact clicks	39,605,099	39,605,099
R02	Fact interaction rows = clickstream rows	10,655,280	10,655,280
R03	`dim_student` = distinct students in `studentInfo`	28,785	28,785
R04/R05	Enrollments / submissions preserved	32,593 / 173,912	32,593 / 173,912
R06	SUM(score) preserved	13,169,342	13,169,342
R07/R08	Both aggregates sum to the fact	39,605,099	39,605,099
R09	Presentations whose clicks disagree with staging	0	0
R13	Enrollment-weeks with both a zero row and clicks	0	0
R16	At-risk label = Fail + Withdrawn count	17,208	17,208

End-to-end rebuild test (M5). The warehouse was dropped and rebuilt from the raw CSVs a second time. That run took 718.9 s and again passed 28 of 28 checks. The leakage check and all model results were then regenerated and are identical to the first run.

5. OLAP

`olap/olap_queries.sql` contains 11 queries covering the operations promised in the proposal. `olap/run_olap.py` runs every variant once to warm up and then five timed runs, and reports the median. Where an aggregate can answer the question, it asserts that the aggregate variant returns exactly the same rows as the base-fact variant.

Table 4: OLAP query pack: median runtimes, base fact vs. materialized aggregate (`out/olap_runtimes.csv`). All 7 aggregate variants return identical rows.

	Operation	Question	Base ms	Agg ms	Speed-up
Q01	roll-up	Clicks by module → presentation, `WITH ROLLUP`	7,460	10.8	688×
Q02	drill-down	FFF: clicks/student by presentation → phase	6,288	10.8	584×
Q03	slice	BBB-2014J: clicks and users by activity group	364	—	—
Q04	dice	BBB, FFF × Oct × weeks 1–4 × 2 groups × outcome	236	—	—
Q05	pivot	Activity type × week 1–10 matrix, DDD-2014J	298	1.9	160×
Q06	cross-tab	Final outcome × week-1–4 engagement decile	2,745	859	3.2×
Q07	cross-term	CCC 2014B vs. 2014J, weekly clicks/student	339	5.2	65×
Q08	roll-up (hier.)	Activity group → type, by outcome	8,935	—	—
Q09	roll-up (fact 2)	Mean score, % late by module → assessment type	357	—	—
Q10	drill-across	Engagement (fact 1) beside scores (fact 2) per presentation	6,731	332	20×
Q11	zero-activity	% of registered students with no clicks, weeks 1–8	1,034	756	1.4×

Aggregate navigation. The speed-up tracks how much the aggregate shrinks the scan. A1 (9,253 rows) is 310 times smaller than the fact, and queries that only need presentation, week and activity run two to three orders of magnitude faster. A2 is only 2.9 times smaller, and its gain on per-student questions (Q06, Q11) is a modest

1.4–3.2×. Questions that split clicks by an enrollment attribute together with an activity attribute (Q04, Q08) cannot use either aggregate. For those questions the base fact remains necessary.

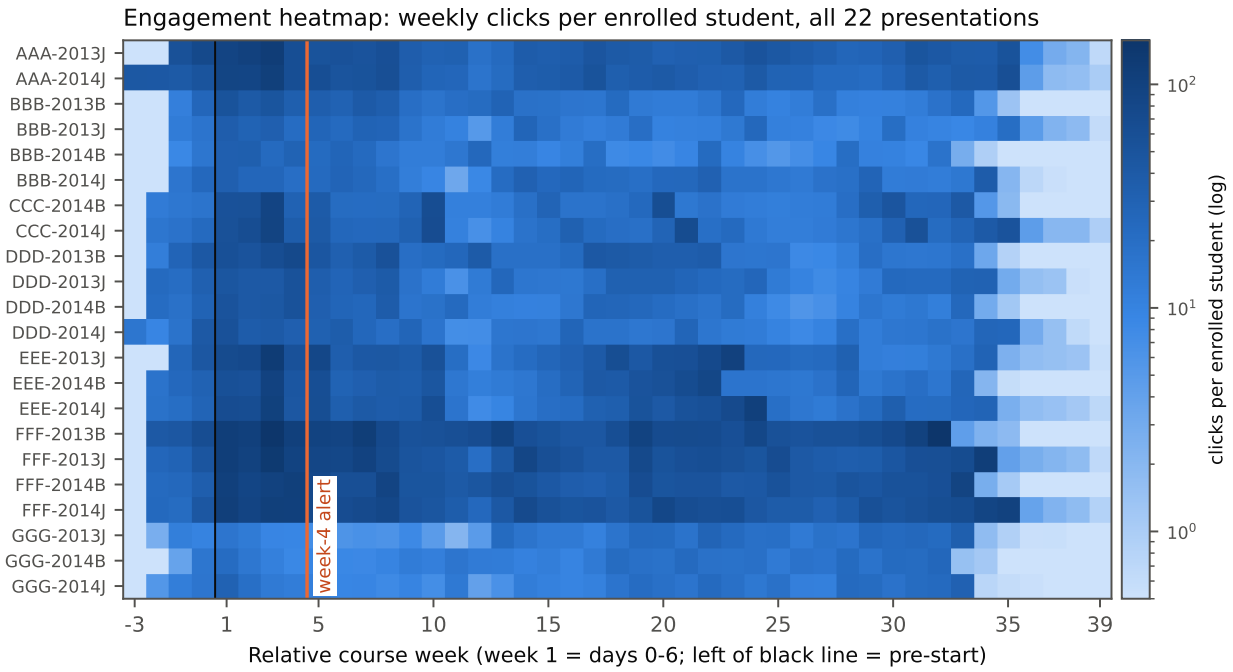


Figure 3: Weekly clicks per enrolled student (A1 aggregate). Every presentation shows a pre-start ramp, peak intensity in the early weeks, sustained mid-course activity, and a sharp drop after its final week (weeks 34–39). FFF and AAA run at roughly 3× the per-student intensity of BBB and GGG.

What the cube shows. Engagement in the first four weeks is strongly ordered against outcome (Q06, Figure **??**). The bottom decile has no week-1–4 clicks at all, and 97% of that decile fails or withdraws. In the top decile 25% do. Already in week 1, 28% of eventual fail/withdraw students who are still registered have zero clicks, against 13% of eventual passers (Q11). By week 8 that gap widens to 42% vs. 14%. The coordinator's motivating question is answered by Q07. CCC's October 2014 cohort engaged more per student than its February cohort in every one of weeks 1–10 (by 2–21%, most in week 4), then less in weeks 11–12. Q10 drills across the two facts. Per-student clicks and mean scores alone do not order presentations by risk: GGG has the lowest engagement but also one of the lowest at-risk rates (38–43%). That is the first sign that *relative*, within-presentation engagement matters more than raw volume, and the models confirm it.

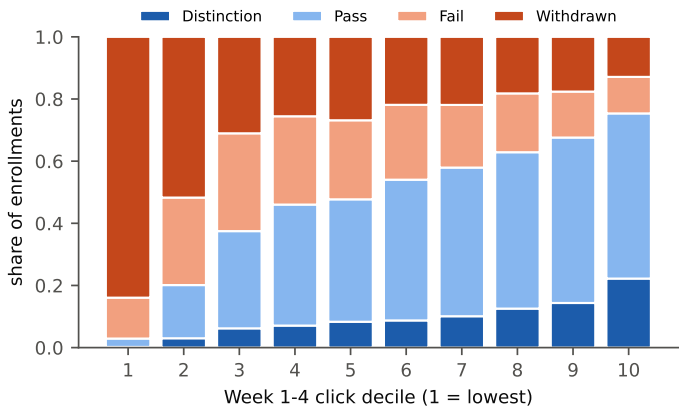


Figure 4: Final outcome by week-1–4 click decile, all 32,593 enrollments (Q06).

6. Mining

6.1 Feature mart and leakage control

`v_features_week4` (`sql/feature_mart.sql`) defines the modeling population and features.

- **Population.** Every enrollment still registered on day 28: 27,553 rows, 44.2% at risk.
- **Features (29).** Registration attributes: prior attempts, studied credits, registration day, and a missing-date flag. Pre-start clicks and active days. Clicks in each of weeks 1–4. Active days, active weeks and last active week in weeks 1–4. Distinct sites in week 4. Clicks per activity group and number of activity types used. Two derived features: clicks relative to the presentation median, and a week-4 vs. week-1 trend. Module indicators.
- **What the view cannot see.** The label, the unregistration date, any assessment data, and every protected attribute. Protected attributes are reachable only through a separate audit view.

The cutoff is enforced in SQL by joining only weeks with `is_early_window = 1`, meaning `last_day` ≤ 27. `mining/check_leakage.py` asserts it independently. The script recomputes all 16 clickstream features from the *raw CSV* with pandas using only days < 28, and they match the SQL view for all 27,553 enrollments. It then repeats the computation with no cutoff, which is what a leaky pipeline would produce, and gets 25,679 mismatches. That shows the test can detect leakage when it is present. Finally, it checks the model's feature list against a forbidden-column list. The result was **LEAKAGE CHECK PASS**.

6.2 Classification

Three models were compared against a majority-class baseline:

- **Logistic regression (LR).** Standardized features, log-transformed counts, and C chosen by inner 3-fold CV.
- **CART decision tree.** Raw counts, with depth and leaf size chosen by inner CV.
- **Gaussian naive Bayes (NB).** Log-transformed counts.

Two validation designs were used. The first is 5-fold stratified CV over all presentations. The second, which matters for deployment, trains on the 10,647 February-start enrollments and tests on the 16,906 October-start enrollments. The decision threshold is never left at 0.5. It is tuned on the *training* out-of-fold probabilities to maximize recall subject to a precision floor of 0.60, then applied unchanged to the test set. The floor reflects the cost asymmetry: an unnecessary check-in costs an advisor minutes, while a missed student may cost a course.

Table 5: 5-fold stratified CV, all presentations (mean $\pm$ s.d., threshold 0.5).

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Brier
Majority baseline	.558 $\pm$ .000	—	.000	.000	.500	.442
Logistic regression	**.673 $\pm$ .003**	**.662 $\pm$ .005**	**.530 $\pm$ .010**	**.589 $\pm$ .006**	**.724 $\pm$ .008**	**.208**
Decision tree (CART)	.652 $\pm$ .003	.634 $\pm$ .010	.505 $\pm$ .039	.561 $\pm$ .022	.689 $\pm$ .009	.218
Naive Bayes	.646 $\pm$ .005	.646 $\pm$ .017	.443 $\pm$ .032	.525 $\pm$ .019	.692 $\pm$ .009	.318

Table 6: Presentation-held-out test (train February $\rightarrow$ test October; test base rate 41.3%).

		Threshold 0.5				Tuned threshold (precision floor 0.60 on train)			
Model	**AUC**	**Acc.**	**Prec.**	**Recall**	t	**Prec.**	**Recall**	**F1**	**Flagged**
Majority baseline	.500	.587	—	.000	—	—	—	—	—
Logistic regression	**.719**	**.668**	.593	.622	.392	.519	**.801**	**.630**	63.7%
Decision tree (CART)	.679	.643	.562	.612	.426	.501	.761	.604	62.6%
Naive Bayes	.692	.652	.628	.382	.001	.523	.734	.611	57.9%

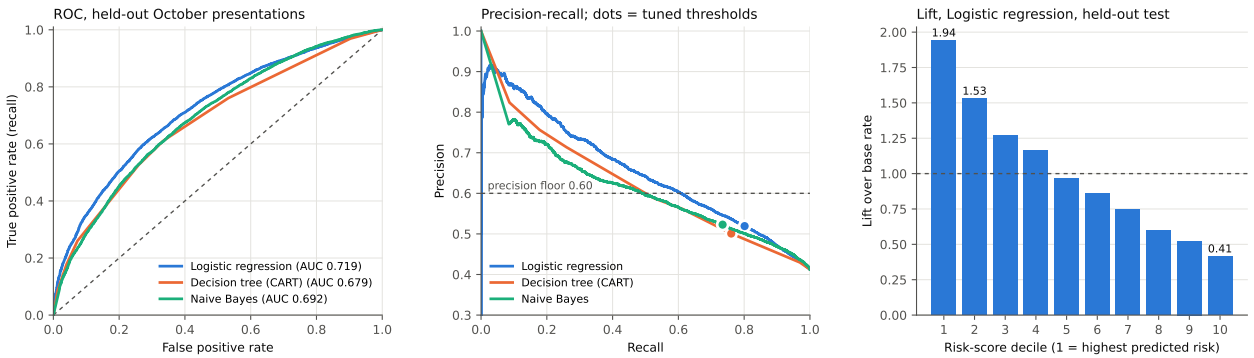


Figure 5: Held-out October test. Left: ROC curves. Centre: precision–recall curves, with dots marking the tuned operating points. Right: lift by risk decile for logistic regression.

Reading the results. Logistic regression is the best model on every summary metric in both validation designs. On the held-out October presentations it beats the majority class by **8.1 accuracy points** (66.8% vs. 58.7%) with AUC 0.719. Its top risk decile is 80.0% at-risk, a lift of 1.94, and the top three deciles contain 47% of all at-risk students. These numbers are modest, and that is the honest result of removing leakage. Four weeks of clicks separate students only moderately well. Three further findings matter for deployment.

- **Prior shift between cohorts.** The at-risk rate is 48.8% in February presentations and 41.3% in October ones. A threshold that gives 0.60 precision on February data gives 0.52 on October data. The floor is therefore a training-time promise, not a guarantee. Table **??** gives the operating menu an advisor would actually choose from.
- **Robustness.** A second split, 2013 $\rightarrow$ 2014, gives AUCs of 0.705 (LR), 0.677 (CART) and 0.691 (NB), with the same ranking. Weka 3.6 on the same Feb/Oct ARFF files reproduces the results (`mining/weka_crosscheck.sh`): Logistic AUC 0.717 (vs. 0.719), J48 0.685 (vs. CART 0.679), and NaiveBayes 0.682 on untransformed counts (vs. 0.692 on log counts).
- **Naive Bayes behaves as predicted.** The nine log-click features have a mean absolute pairwise correlation of 0.57, and 7 of 36 pairs exceed 0.7. NB therefore double-counts the same evidence. Its ranking (AUC) stays competitive, but its probabilities are badly calibrated (Brier 0.320 vs. 0.212), so its tuned threshold collapses to 0.001. NB is usable as a ranker but not as a probability.

Table 7: Operating-point menu for logistic regression: thresholds chosen on training out-of-fold probabilities, evaluated on the October test.

Precision floor (train)	Threshold	Test precision	Test recall	Test F1	Students flagged
0.55	.312	.468	.905	.617	79.7%
0.60	**.392**	**.519**	**.801**	**.630**	**63.7%**
0.65	.461	.565	.688	.620	50.3%
0.70	.540	.620	.549	.582	36.5%

Interpretability. The CART grid search chose depth 3, and 93% of its split importance is on one feature: week-1–4 clicks *relative to the presentation median*. The depth-3 tree reads as three alert rules:

- Below the presentation median by more than about 30% (log-ratio ≤ -0.34): flag.
- Near the median but active in at most 3 of the 4 weeks: flag.
- Above the median: do not flag.

Logistic regression tells the same story with more nuance. The largest standardized coefficients that reduce risk are active days (–0.25), relative clicks (–0.20), last active week (–0.17), week-2 clicks (–0.13) and pre-start clicks (–0.13). Two features increase risk: prior attempts (+0.11), and navigation-only (homepage) clicks (+0.12), which can be read as logging in without opening content. Module effects are as large as any behavior (GGG –0.31, BBB –0.20, DDD +0.17). Note that module AAA has no February presentation, so the held-out model never saw it. An early version also included raw total clicks, which made the LR coefficients collinear (+1.19 total vs. –1.11 relative). It was removed for interpretability, and AUC was unchanged (0.716 $\rightarrow$ 0.719).

6.3 Clustering: engagement archetypes

k-means ($n_{\text{init}} = 10$) was run for $k = 2..10$ on nine standardized week-1–4 features: log clicks for the pre-start period and each of weeks 1–4, active days, assessment and communication shares, and activity types. Outcomes were not inputs. Mean silhouette peaks at $k = 2$ (0.306), but $k = 2$ is only a high/low volume split. The elbow is at $k = 4$: within-cluster SS falls 15% from 3 to 4 clusters and only 7% from 4 to 5, while silhouette holds at 0.248 before dropping to 0.195 at $k = 5$. $k = 4$ was therefore chosen. Ward hierarchical clustering on a stratified sample of 4,000 enrollments, cut at 4, agrees with k-means at adjusted Rand index **0.64**. For comparison the ARI is 0.22 at $k = 2$, so the four-cluster structure is the more reproducible one.

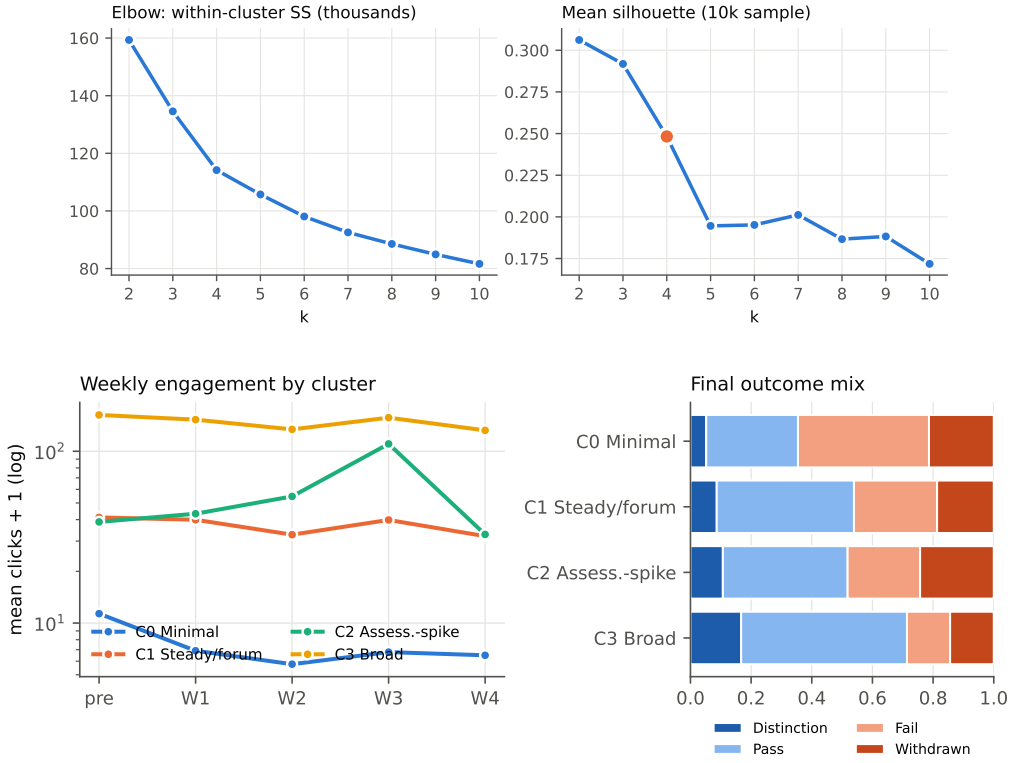


Figure 6: Top: model selection. Bottom: mean weekly clicks per cluster (log scale) and final-outcome mix.

Table 8: Engagement archetypes, $k = 4$ (27,553 enrollments still registered at week 4; `out/cluster_profiles.csv`).

Archetype	Share	Median clicks	Active days	Assess. share	Forum share	Withdrawn	At risk
C0 Minimal presence	19.6%	14	2.2	1%	7%	21.4%	**64.6%**
C1 Steady, forum-engaged	32.2%	120	8.4	2%	31%	18.7%	46.1%
C2 Assessment-spike	14.7%	205	8.8	54%	7%	**24.3%**	48.2%
C3 Broad and consistent	33.5%	466	18.6	11%	33%	14.4%	**28.6%**

The archetypes are substantive, not just volume bands. C2 averages only about 32–54 clicks in weeks 1, 2 and 4, then spikes to 109 in week 3 when the first quiz/CMA falls due, and 54% of its clicks are on assessment activities. It is the “assessment-only” pattern the proposal anticipated, and it has the highest withdrawal rate (24%) despite higher volume than C1. The proposal’s hypothetical “front-loaded then absent” archetype did *not* emerge as a cluster in weeks 1–4. Those students are spread across C0 and C1, and separating them would need later weeks, which the alert cutoff excludes. Of the C0 students who were still registered at week 4, 65% go on to fail or withdraw. That makes C0 the obvious first call list. C3 has the highest distinction rate (16.7%).

6.4 Disparate-impact audit

After prediction, the October test predictions of the deployed model (LR at $t = 0.392$) were joined to the protected attributes and summarized in aggregate (`mining/audit.py`; Table **??**). For each group the audit reports how often students are flagged, how often they are truly at risk, and the recall and false-positive rate. The screen applied is the four-fifths rule on flag rate: lowest group divided by highest, among groups with $n \geq 100$.

Table 9: Disparate-impact audit, held-out October test (16,906 enrollments).

Attribute	Group	n	Actual at-risk	Flag rate	Precision	Recall	FPR
Gender	F	7,492	.397	.622	.494	.775	.522
	M	9,414	.425	.648	.539	.821	.520
Age band	0–35	11,685	.430	.678	.523	.825	.567
	35–55	5,076	.375	.548	.508	.743	.431
	55+	145	.303	.407	.492	.659	.297
IMD quintile	0–20% (most deprived)	3,278	.508	.656	.614	.793	.515
	20–40%	3,603	.445	.646	.557	.809	.516
	40–60%	3,374	.407	.637	.509	.797	.527
	60–80%	3,155	.375	.631	.485	.815	.521
	80–100% (least)	2,879	.337	.627	.435	.808	.535
Disability	N	15,328	.401	.634	.508	.802	.522
	Y	1,578	.520	.658	.629	.795	.509

Findings, reported whatever they show.

- **Gender and disability** pass comfortably: flag-rate ratios of 0.96 and 0.96, recall gaps of at most 4.7 points, and FPR gaps under 1.4 points.
- **Age band fails the screen.** The flag-rate ratio is 0.60, driven by the small 55+ group. Even between 0–35 and 35–55 the ratio is 0.81, right at the line. Younger students are flagged more, and their false-positive rate is 13.6 points higher than the 35–55 group (0.567 vs. 0.431). Their actual risk is also higher, but the model over-flags them relative to that risk (flag/actual 1.58 vs. 1.46).
- **IMD reveals a subtler pattern.** Flag rates are almost flat (0.63–0.66, ratio 0.83, passing the screen) even though actual risk falls from 51% to 34% across quintiles. Recall is equal across groups. The consequence is that a flag means less for affluent students (precision 0.44) than for deprived ones (0.61). Because the model never sees IMD, it cannot reproduce that gradient. Whether that is the desired form of fairness is a policy decision rather than a modeling one.

Any deployment should monitor the age-band FPR gap, and should present a flag as an invitation to check in, never as a judgment.

7. LLM component: natural-language queries over the warehouse

Design. An open-weight code model writes MySQL for 40 held-out analyst questions. The gold SQL comes from the OLAP pack and variants of it (`text2sql/questions_test.py`). Each question states the columns it expects, so "same rows" is well defined. Matching is order- and column-order-insensitive, with numbers compared at two decimals. 22 questions can be answered exactly from an aggregate table, and they define the *aggregate-navigation rate*. The three conditions are:

- **No schema.** The model gets only the dialect and the question.
- **Schema in prompt.** The context is built *programmatically from the live warehouse* (script `schema_context.py`): compact DDL with keys and foreign keys for the 10 dimension, fact and aggregate tables, each table's grain statement read back from its comment, and a data dictionary of every low-cardinality value. With the instructions, the prompt is 2,438 tokens. Staging tables are deliberately excluded.
- **LoRA + schema.** The same prompt, after LoRA fine-tuning [5] on 200 (question, SQL) pairs. The pairs come from 21 parameterized templates (`text2sql/make_train_pairs.py`). Each pair was executed and checked to return rows, and none duplicates a test question or its gold SQL.

What ran where. The proposal's degradation path was used for the prompt-only conditions. Qwen2.5-Coder-7B-Instruct [6], the proposal's first-choice model, ran as a 4-bit GGUF (Q4_K_M) through `llama.cpp` on the 2-vCPU build machine, with greedy decoding. The 1.5B sibling ran the same way as a size contrast. The LoRA condition needs a GPU. It is packaged for the course's NRP hub (`text2sql/nrp/`): training script, the identical prompt, a generate-only evaluator, and a scorer that runs back on the warehouse machine. Its rows in Table **??** are filled from that run. No LoRA number is reported until it exists.

Table 10: Text-to-SQL results on the 40 held-out questions (22 aggregate-eligible). Execution match = same multiset of rows as the gold query; aggregate navigation = share of aggregate-eligible questions answered from an `agg_` table.

Model	Condition	Run	Exec. match	Invalid SQL	Agg. navigation	s/q
Qwen2.5-Coder-7B (4-bit)	no schema	CPU	0.0% (0/40)	100.0%	0.0%	29
Qwen2.5-Coder-7B (4-bit)	schema in prompt	CPU	47.5% (19/40)	22.5%	90.9%	57
Qwen2.5-Coder-1.5B (4-bit)	no schema	CPU	0.0% (0/40)	100.0%	0.0%	6
Qwen2.5-Coder-1.5B (4-bit)	schema in prompt	CPU	15.0% (6/40)	75.0%	81.8%	12
Qwen2.5-Coder-1.5B (fp16)	no schema	NRP GPU		*pending NRP run*		
Qwen2.5-Coder-1.5B (fp16)	schema in prompt	NRP GPU		*pending NRP run*		
Qwen2.5-Coder-1.5B (fp16)	LoRA + schema	NRP GPU		*pending NRP run*		

Results. Schema context is decisive. Without it, neither model produced a single executable query. Every attempt invented plausible tables (`oulad_dw.vle`, `students`, `enrollments`). With the warehouse-generated context, the 7B model matched **19 of 40** questions (47.5%), and only 9 of its queries failed to execute. The 1.5B model matched 6 of 40, and 75% of its SQL was invalid. On aggregate navigation, the 7B model reached for an `agg_` table on 20 of the 22 eligible questions (**91%**), scanned the base fact on only 2, and every one of its 8 correct aggregate-eligible answers used an aggregate. The grain comments make the aggregates *discoverable*: the model knows they exist and when to prefer them. Discoverable is not the same as *usable*, though. The 7B model was more accurate on the questions that did *not* involve aggregates (61%, 11/18) than on those that did (36%, 8/22).

Where the design was not legible. Each of the 7B model's 21 failures was classified by primary cause.

- **Attribute on the wrong table (9; all of the invalid SQL).** For example, `code_presentation` on an aggregate, `highest_education` on the enrollment, or `pres_year` on the enrollment. The model assumes denormalized attributes that a strict star schema keeps in the dimensions.
- **Surrogate key used as a value (4).** For example, `week_key = 2` and `week_key <= 0`. The surrogate key is offset from `week_num` by 4, so these queries run but return the wrong week.
- **Composite code in the wrong column (2 primary, 7 overall).** For example, `code_presentation = 'CCC-2014J'` where `module_presentation` was meant.
- **Wrong grain in the arithmetic (6).** Averaging over enrollment-*weeks* instead of students, and in one case joining the presentation-level aggregate to enrollments, which multiplies the clicks.

The second and fourth causes are specific to dimensional design. Integer surrogate keys and pre-aggregated grains are exactly what make the warehouse fast and correct for its designer, and exactly what an outside reader misreads. The result suggests three inexpensive design changes:

- Carry `week_num` and `module_presentation` as degenerate attributes on the aggregates.
- State in the dictionary that `week_key` is not a week number.
- Include one worked aggregate query in the context.

Testing those changes is future work. The LoRA condition tests the other route, which is to teach the model the schema's conventions rather than to change the schema.

8. Deviations from the proposal

- **Fact 1 grain refined.** The grain gained the activity-type dimension plus one zero-activity member (Section 3), which keeps the zero-row promise without 20× padding.
- **Modeling population.** 5,040 enrollments that left before day 28 were excluded (D13). The proposal did not specify this. It is the main reason the accuracy figures are lower than published whole-course figures.
- **Precision floor.** The proposal left the floor unstated. 0.60 was chosen, and the full menu is in Table **??**.

- **Feature change.** Raw total clicks were dropped from the models for collinearity, as disclosed above.
- **Data source.** The data came from a mirror, with reconciliation (D1).
- **Compute.** The project was built and timed in a Linux container (2 vCPU, 8 GB) rather than on the Windows workstation. The code is portable, and connection settings come from environment variables.
- **LLM component.** The two prompt-only conditions ran on CPU with a 4-bit quantized model rather than on NRP. The LoRA condition is packaged for NRP and its results are pending (Section **??**).

9. Conclusion

Five of the proposal's six success criteria are met in full, and the sixth in part:

- The warehouse rebuilds with one command and passes every reconciliation assertion.
- All OLAP queries return verified results, with measured aggregate speed-ups of 1.4–688$\times$.
- Three classifiers are compared under a machine-verified week-4 cutoff, with CV standard deviations, a lift chart, a recall-oriented tuned threshold, and an 8.1-point margin over the baseline.
- $k = 4$ is defensible and substantively characterized.
- The audit is reported, including the criterion it fails.
- *In part:* the text-to-SQL evaluation reports execution match for two of the three conditions (no schema, schema in prompt). The LoRA condition awaits its NRP run.

The substantive finding is narrower than the headline numbers often quoted for OULAD. Four weeks of behavior, measured *relative to classmates*, identifies about four in five students who will fail or withdraw, at the cost of flagging about two-thirds of the class. The highest-risk fifth of the class (archetype C0) is visible from almost no data at all. For an advisor, the practical output is the three-rule tree plus the C0 list, not the AUC.

AI use

This project was AI-assisted, as the proposal planned. Claude (Anthropic) wrote and ran the pipeline code, the SQL, the analyses, and the first draft of this report under my direction. The itemized log in `AI_USE_LOG.md` lists each task and what I reviewed and verified.

References

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